MIL-C-24643/55B
3 October 1986
SUPERSEDING
MIL-C-0024643/55A(SH)
8 November 1985
MIL-C-24643/55
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS4NW8 (INCLUDING VARIATION LS4NWA8)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LS4NW8 (unarmored) and LS4NWA8 (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated, AWG no. 8, class B stranding, (7 by 0.0486 inch).
Second - Separator (white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene or ethylene propylene rubber insulation, colored black. Nominal thickness 0.045 inch. Standard identification code applied by method 1.
Fourth - Four conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be employed as necessary to form a firm, well-rounded assembly.
Fifth - Binder tape applied helically with overlap.

Construction variant for type LS4NW8

Sixth - Cross-linked polyolefin jacket. Nominal thickness of 0.050 inch. Maximum overall diameter 0.74 inch. Cable surface marking required.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LS4NWA8

(B) Sixth - Cross-linked polyolefin jacket. Nominal thickness of 0.050 inch. Maximum overall diameter 0.74 inch. Cable surface marking not required.
Seventh - Braided metal armor and paint.

NOTES: 1. Variant LS4NW8 military part number - M24643/55-01UN.

2. Variant LS4NWA8 military part number - M24643/55-02AN.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	0.679
Voltage withstand - conductor to conductor, volts, root mean square, minimum.....	4500
Insulation resistance - megohms/1000 feet, minimum.....	4200
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Group B:	
Cold bending, cable - 6-inch mandrel diameter, maximum..	No damage
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Cross-linked polyethylene	
Tensile strength - lbf/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Ethylene propylene rubber	
Tensile strength - lb/in, minimum.....	700
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lbf/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Cross-linked polyethylene	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum.....	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60

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EXAMINATION AND TESTS: (Continued)

	Requirements
Insulation	
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (conductor) - cycles, minimum	25
Permanence of printing (jacket) type LS4NW8 only - cycles, minimum	250
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	Zero
Armor - conformance to materials, construction and coverage (type LS4NWA8 only)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cold working (minus 20 ± 2°C)	No damage
Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

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UNIT ORDERING LENGTH: 1500 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-55)

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